MULTI-SHIELD — Development Master Plan

Full Title: MULTI-SHIELD: Failure-Aware Decision Policies for Reliable Multimodal AI under Compound Degradation
Domain: Artificial Intelligence & Machine Learning
Research Area: Robust Multimodal Learning · Trustworthy AI · Uncertainty Estimation · Selective Prediction
Sources: Master Project Blueprint (56 pages) · Master Project Plan (31 pages)

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1. What the Project Is

MULTI-SHIELD is a **reliability-aware multimodal AI architecture** that investigates how an AI system can:

1. **Identify** unreliable, missing, noisy, contradictory, or potentially manipulated modalities
2. **Dynamically decide** how much each modality should influence the final decision
3. **Abstain** entirely when evidence is collectively insufficient

What MULTI-SHIELD is NOT

■ NOT This	■ But This
Another multimodal classifier	A **reliability diagnosis and decision framework**
Another sentiment-analysis model	An investigation into **evidence management before prediction**
Simple missing-modality imputation	**Compound failure diagnosis and adaptive intervention**
Simple attention-based fusion	**Failure-state-conditioned policy routing**
A generic "trust score"	**Multi-dimensional reliability state** (confidence + uncertainty + agreement + quality + failure type)

> **IMPORTANT:** The single sentence to remember: *"MULTI-SHIELD investigates whether multimodal AI can become more reliable by explicitly diagnosing the reliability and failure state of heterogeneous evidence before deciding how that evidence should influence the final prediction."*

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2. What It Is Focusing On

The candidate research direction is:

> **Reliability diagnosis and decision-making under compound cross-modal failure.**

The central question:

> *Can a multimodal AI system distinguish different causes of cross-modal unreliability and dynamically alter its fusion and decision policy according to the diagnosed reliability state?*

The Key Word: **Diagnose**

Not simply:

> *"This modality has weight 0.2."*

But:

> *"Why is its reliability low? Is it missing? Noisy? Contradicting other evidence? Or just providing complementary but different information?"*

Five Innovation Layers Under Investigation

Layer	Description	Status
A. Reliability Diagnosis	Determine *why* a modality is unreliable	Core hypothesis
B. Failure-Aware Fusion	Fusion policy changes per diagnosed failure state	Core mechanism
C. Compound-Failure Handling	Evaluate combinations, not just isolated failures	Key differentiator
D. Selective Decision (Abstention)	System can abstain when evidence is insufficient	Safety feature
E. Reliability Benchmark (MS-Bench)	Controlled benchmark with multiple failure types/severities	Potential contribution

> **NOTE:** We cannot claim all five as novel. The literature review determines which becomes the actual contribution.

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3. What Is the Cause (The Problem)

The Assumption That Fails

Modern AI increasingly combines text, images, audio, video, sensor streams, and structured data. The assumption is:

\text{More modalities} \rightarrow \text{More information} \rightarrow \text{Better decisions}

This assumption fails when one or more modalities are:

Failure Type	Example	Severity
Missing	Camera feed drops, audio unavailable	Common in deployment
Noisy	Audio contains heavy interference	Gradual degradation
Corrupted	Input technically present but information is damaged	Hard to detect
Contradictory	Image → DOG, Text → CAT	Ambiguous without context
Misleading/Manipulated	Intentionally altered modality	Adversarial scenario
Semantically Ambiguous	Individually valid but describe different aspects	Requires contextual reasoning

Collectively Insufficient	All available evidence is weak	Requires abstention

The Four Root Causes (from Project Plan)



Why It Matters (Evidence from Literature)

- **ICLR 2025**: Foundation models recognize conflicts in unimodal contexts far more reliably than across modalities (cross-modal attention imbalance)
- **ICML 2025**: Modality conflict formalized as a source of hallucination (MMMC dataset)
- **AISTATS 2025**: Uncertainty estimation becomes particularly difficult when modalities conflict
- **2026 Survey**: Organizes reliability challenges into overlapping categories with unresolved questions in low-quality multimodal learning

> **WARNING: The Most Dangerous Scientific Mistake:** \text{Cross-Modal Disagreement} \neq \text{Unreliability} *Example: Image → person smiling, Text → "I am sad" — both can be valid (sarcasm, suppressed emotion). Disagreement does not automatically mean one modality is wrong.*

> The system must distinguish: **True contradiction** vs. **Complementary information** vs. **Ambiguity** vs. **Noise** vs. **Missing evidence**

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4. How We Intend to Resolve It

The Research Hypothesis

> *A multimodal system that explicitly diagnoses modality reliability and conditions its fusion/abstention policy on that diagnosis will be more robust under compound modality degradation than conventional and recent adaptive-fusion approaches.*

The Three-Part Decision Process



The Four Candidate Actions

Action	When to Use	Risk
REWEIGHT — use less, not none	Modality remains informative but less reliable	May still include noisy signal
SUPPRESS — exclude entirely	Modality is likely to harm the decision	May discard valid complementary info
REPAIR — attempt recovery	Degraded representation reconstructable from other modalities	Reconstructed features may hallucinate
DEFER — decline prediction	Remaining information doesn't justify a reliable decision	Reduces coverage/throughput

> **TIP: Development Strategy:** Start with `{REWEIGHT, SUPPRESS, DEFER}` in Phase 1. Introduce `REPAIR` only after the baseline policy is proven stable. REPAIR is technically high-risk — synthesized features can inject hallucinations.

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5. What Is Our Approach

The MULTI-SHIELD Approach (Failure-State-Conditioned Policy Routing)



The research question: *Does knowing the type of failure improve the choice of intervention?*

How This Differs from Normal ML

Paradigm	Process	Limitation
Normal ML	Input → Model → Prediction	No evidence assessment
Conventional Multimodal	Image + Audio + Text → Fusion → Prediction	Assumes all inputs useful
Adaptive Weighting	Inputs → Estimate reliability → Calculate continuous weights → Fuse → Predict	Doesn't distinguish *why* reliability is low
MULTI-SHIELD	Inputs → Diagnose *what's wrong* → Select appropriate intervention → Fuse adaptively → Predict/Abstain	Evidence management *before* decision-making

The Important Distinction

> We are **NOT claiming** that existing methods literally cannot suppress or abstain. Some already support related capabilities.

> Our question is narrower: **Does explicitly modelling failure states and conditioning intervention selection on those states outperform a strong, fairly tuned continuous-weighting baseline under compound degradation?**

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6. What Are Other Approaches

Comprehensive Landscape of Existing Approaches

#	Approach	How It Works	Key Methods	Limitation We Investigate
1	**Fixed Fusion** (Early/Late/Attention)	Combines all modalities using fixed architecture	Tensor Fusion Network, early concat, late averaging	Cannot adapt to sample-specific failures
2	**Adaptive Weighting**	Changes each modality's contribution according to estimated reliability	DMS (2025), attention-based weighting	Does continuous weighting suffice for severe/interacting failures?
3	**Missing-Modality Learning**	Trains model to handle absent inputs	Modality dropout, prompt learning, MCCF (2026)	Doesn't automatically solve noise, conflict, and corruption together
4	**Uncertainty-Aware Fusion**	Uses uncertainty estimates to influence predictions	Evidential Deep Learning, Discounted Belief Fusion (AISTATS 2025)	Uncertainty alone may not identify *which intervention* is appropriate
5	**Restoration/Imputation**	Reconstructs missing or damaged information	PRIME (2026), autoencoders, cross-modal generation	Reconstructed features may be inaccurate or misleading
6	**Conflict-Aware Methods**	Detects and handles cross-modal disagreement	MMMC (ICML 2025), conflict-aware attention	May not distinguish valid divergence from corruption

7	**Selective Prediction**	Abstains when prediction risk too high	Conformal prediction, risk-coverage methods	Must be integrated with reliability management
8	**Sensor-Failure Robustness**	Expert-based handling of sensor dropout	Multi-Modal Expert Fusion (CVPR 2025)	Domain-specific, doesn't generalize failure taxonomy

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7. Why Is Our Approach Better

Three Hypothesized Advantages

Advantage	Explanation	How We Test It
1. More Appropriate Intervention	The system learns that *different conditions* may favour *different actions* — a missing modality needs SUPPRESS, a noisy one needs REWEIGHT, a contradictory one needs conflict-specific handling	Compare action-selection quality vs. uniform weighting
2. Better Compound-Failure Handling	The policy considers multiple degraded modalities *jointly* rather than treating each failure in isolation	Stress-test with 2-way and 3-way compound degradation
3. More Responsible Decisions	The system can DEFER when evidence is inadequate rather than forcing a prediction	Measure selective risk (AURC) and risk-coverage trade-offs

Honest Acknowledgement of Disadvantages

Disadvantage	Reality
Complexity	A policy router adds architectural complexity
Cascading Error	Misdiagnosed failure states propagate to wrong interventions
Repair Risk	Cross-modal reconstruction can introduce hallucinations
Overhead	Multiple diagnostic signals + policy evaluation increases latency

> **IMPORTANT:** We must **measure these costs** rather than assume the architecture is superior. If a strong scalar baseline performs equally well, the central hypothesis is not supported — and that is a valid research result, not a failure.

8. Already-Present Approaches & How We Differ

Detailed Prior Art Comparison

Existing Method	What It Does	What It Doesn't Do	How MULTI-SHIELD Differs
PRIME (2026)	Explicit reliability diagnosis → prototype-conditioned restoration → reassessment → reliability-weighted fusion	Doesn't model multiple *intervention types* as a policy choice; focuses on intent recognition	We investigate whether selecting among {Reweight, Suppress, Repair, Defer} based on failure state outperforms a single restoration + reweighting pipeline
DMS (2025)	Adaptive per-sample modality weighting via confidence, uncertainty, and semantic consistency	Uses *continuous scalar weights* — doesn't diagnose *why* reliability is low or offer discrete interventions	We condition the intervention on the *type* of failure, not just its magnitude
MCCF (2026)	Conformal fusion under missing modalities with calibrated uncertainty	Focused specifically on *missing* modalities; doesn't address noise, conflict, or compound failure	We extend beyond missingness to a full failure taxonomy under compound degradation
MMMC (ICML 2025)	Modality conflict formalized as hallucination source; benchmark dataset	Identifies conflict but doesn't offer a *decision policy* for handling it	We integrate conflict detection into a broader diagnosis-and-action framework
AISTATS 2025 (Belief Fusion)	Conflict-aware uncertainty estimation via evidential deep learning	Provides better uncertainty but doesn't route to different interventions	We use uncertainty as one *input signal* to a learned policy router
CVPR 2025 (Expert Fusion)	Multi-modal expert fusion under sensor failures for autonomous driving	Domain-specific (LiDAR + camera); doesn't generalize failure taxonomy	We propose a domain-agnostic failure diagnosis that works across text/audio/image

The Key Differentiator



> The shift is from **"how much to trust"** (scalar) to **"what went wrong and what to do about it"** (diagnostic + policy).

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9. The Core of Our Solution

The core is NOT the dashboard, NOT the module names, NOT the four action labels.

The Core Is:

> **The experimental demonstration that failure-state-conditioned intervention selection provides a measurable advantage over existing alternatives.**

Three Levels of Contribution

Level	Contribution	Description
Primary	**Failure-Aware Decision Policy**	Test whether a multi-class failure-state representation improves intervention selection under interacting degradation
Supporting	**Reproducible Compound-Degradation Evaluation Framework**	Evaluate missingness, noise, corruption, conflict and combinations using controlled severity, repeatable splits, documented metadata
Practical	**Explainable Reliability Dashboard**	Show detected condition, chosen action, confidence, latency, and outcome relative to baseline

The Strongest Possible Research Narrative

> *"Existing multimodal systems typically treat reliability as an implicit weighting problem. MULTI-SHIELD investigates reliability as an explicit diagnostic state that informs fusion and decision-making under compound cross-modal degradation."*

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10. Target Users

MULTI-SHIELD is a **research framework**, not yet a standalone product. Its target users depend on the application:

User Group	Use Case	Example Scenario

Assistive-Technology Developers	Systems combining camera + speech + text that detect when evidence is inadequate	A visual assistance app that refuses to describe a scene when camera feed is corrupted rather than hallucinating
Autonomous Systems Researchers	Perception systems responding to sensor dropout, occlusion, conflicting measurements	Self-driving module that shifts trust from a foggy camera to LiDAR + radar
Educational AI Developers	Classroom assistants coping with noisy speech, incomplete lectures, unreliable OCR	Lecture summarizer that flags low-confidence audio segments instead of transcribing noise
Industrial Monitoring Teams	Reliable decisions when different sensors provide degraded/inconsistent evidence	Factory monitor that escalates when vibration sensor contradicts temperature readings
Disaster Management Systems	Satellite + sensor + field reports with contradictory information	Emergency system that flags conflicting ground reports vs. satellite imagery

> **TIP: Recommended Strategy:** Use a multimodal sentiment/emotion benchmark (CMU-MOSI/MOSEI/MELD) as the controlled research environment. Build the demo around an assistive or monitoring scenario only if it can be validated without unsupported safety claims.

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11. What It Targets

Primary Research Target

The **compound degradation problem** — the scenario where multiple inputs fail simultaneously in different ways:



The Six Evaluation Dimensions

Dimension	Metrics	Purpose
Prediction Quality	Accuracy, Precision, Recall, F1, AUROC	Does it still get the right answer?
Reliability/Calibration	ECE, Brier Score, calibration curves	Does confidence match reality?
Failure Diagnosis	Failure classification F1, severity estimation error	Can it diagnose what went wrong?

Robustness	Performance degradation, relative robustness, area under degradation curve	How gracefully does it degrade?
Selective Decision	Coverage, selective risk, AURC, risk-coverage curve	Does it know when to abstain?
Efficiency	Inference latency, memory, parameter count, computational overhead	What does the policy cost?

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12. How the Architecture Works

Full 5-Layer Architecture

Layer-by-Layer Breakdown

Layer 1 — Modality Encoders

- **Purpose:** Convert each raw input into numerical feature representations
- **Implementation:** Frozen pretrained encoders (no giant foundation model training)
- **Models:** CLIP (image), BERT/RoBERTa (text), Audio Spectrogram Transformer / Wav2Vec (audio)
- **Key Decision:** Encoders are **frozen** — we operate on extracted feature embeddings, keeping training lightweight

Layer 2 — Evidence Assessment

- **Purpose:** Calculate diagnostic signals for each modality
- **Signals Computed:**
- Predictive entropy (how uncertain is this modality's prediction?)
- Epistemic uncertainty (is this out-of-distribution?)
- Feature-quality indicators (signal-to-noise, completeness)
- Cross-modal agreement (do modalities tell the same story?)
- **Critical Rule:** Each signal must be tested to verify it adds useful information

Layer 3 — Failure-State Estimation

- **Purpose:** Classify the likely condition of each modality
- **Output:** Probabilistic estimate over failure taxonomy: `{healthy, missing, noisy, corrupted, contradictory}`
- **Requirement:** Must output **uncertainty about the diagnosis itself** — cascading diagnosis error is a top risk
- **Implementation:** Classifier or probabilistic model trained on controlled degradation data

Layer 4 — Policy Router

- **Purpose:** Select an intervention based on the estimated failure state and expected utility
- **Actions:** `{REWEIGHT, SUPPRESS, REPAIR, DEFER}`
- **Training:** Learned from training data only — **never** supplied with ground-truth corruption labels at inference time
- **Key Constraint:** The policy must be evaluated fairly against a tuned scalar baseline

Layer 5 — Decision and Audit Output

- **Purpose:** Make prediction or defer; output full audit trail
- **Output Fields:**
- Prediction class (or abstention)
- Confidence score
- Action taken per modality
- Reason for action
- Inference latency
- Comparison with baseline decision

The Reliability Engine (Where Novelty Lives)



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13. References

Core Prior-Art References

#	Paper	Year	Venue	Relevance
1	**PRIME** — Adaptive Modality Reliability Diagnosis and Restoration for Robust Multimodal Intent Recognition	2026	arXiv	Closest competitor: explicit diagnosis, restoration, reassessment, reliability-weighted fusion
2	**DMS** — Dynamic Modality Scheduling via Confidence, Uncertainty, and Semantic Consistency	2025	arXiv	Adaptive per-sample weighting; strong scalar-weighting baseline

3	**MCCF** — Conformal Fusion Under Missing Modalities	2026	arXiv	Missing-modality robustness, evidential fusion, conformal calibration
4	**Multimodal Fusion on Low-Quality Data: A Comprehensive Survey**	2026	Information Fusion	Taxonomy of noisy, incomplete, quality-varying multimodal data
5	**Deep Multimodal Learning with Missing Modality: A Survey**	2024	arXiv	Breadth of missing-modality research
6	**Resilient Sensor Fusion via Multi-Modal Expert Fusion**	2025	CVPR	Sensor-failure robustness in autonomous driving
7	**Benchmarking Multi-modal Segmentation under Sensor Failures**	2025	CVPR Workshops	Controlled missing/noisy modality evaluation
8	**MMMC** — Robust MLLMs Against Modality Conflict	2025	ICML	Modality conflict → hallucination; MMMC benchmark
9	**Cross-Modal Attention Imbalance** — Why Foundation Models Struggle	2025	OpenReview	Foundation model failure under distributed conflicts
10	**Discounted Belief Fusion** — Multimodal Uncertainty Quantification	2025	AISTATS	Conflict-aware uncertainty estimation

Dataset & Tooling References

Resource	Description	Use
CMU-MOSI	2,199 opinion-video clips with sentiment labels	Small, controlled initial experiments

CMU-MOSEI	Larger collection with sentiment and emotion annotations	Main robustness experiments
MELD	Multimodal dialogue with emotion/sentiment annotations	Cross-dataset transfer testing
MMMC	Modality-conflict benchmark	Conflict-focused experiments
CMU Multimodal SDK	Dataset access and processing toolkit	Data pipeline foundation

Technology Stack

Layer	Tools
Language	Python
Framework	PyTorch
Models	Hugging Face Transformers
Vision	OpenCV
Audio	librosa / torchaudio
Data	NumPy, Pandas, scikit-learn
Visualization	matplotlib, Plotly
Dashboard	Streamlit or Next.js

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14. Our Greatest Contribution

Not the interface. Not the module names. Not the four action labels.

The greatest possible result is:

> Demonstrating on unseen compound-degradation conditions that: 1. MULTI-SHIELD achieves **lower selective risk** at comparable coverage 2. It maintains **competitive Macro-F1** on clean data 3. It chooses interventions **more effectively** than a tuned scalar baseline 4. The improvement **transfers to another dataset** 5. Its additional computation is **justified by the benefit**

If We Also Solve This, It Becomes Exceptional:

> **Distinguishing valid semantic divergence from genuine evidence failure.**

> This becomes a research contribution ONLY if we can: - Define it rigorously - Label it reliably - Demonstrate that the distinction improves decisions

The Four Critical Distinctions

Concept	Definition	NOT the same as:
Reliability	How trustworthy is this modality for the current decision?	Confidence
Confidence	How confident is the model in its prediction?	Reliability
Uncertainty	How much uncertainty exists around the prediction?	Disagreement
Agreement	Do modalities provide consistent evidence?	Reliability

> Explicitly separating these is where the project becomes substantially more sophisticated than simple attention-weighting.

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15. Impact

Six Impact Dimensions

#	Impact Area	Description
1	**Safety**	A system should not confidently act on corrupted evidence
2	**Reliability**	Gracefully degrade rather than catastrophically fail
3	**Explainability**	Instead of "Confidence = 0.87", provide: "Image and audio agree; text conflicts; text reliability reduced"
4	**Robustness**	Performance becomes less sensitive to individual modality failure
5	**Operational Efficiency**	System can avoid processing/trusting unreliable streams unnecessarily
6	**Research Infrastructure**	A controlled framework for studying multimodal failure

SDG Alignment

SDG	Alignment	Condition
SDG 9 (Industry, Innovation, Infrastructure)	Strongest general alignment	Demonstrated through research output
SDG 10 (Reduced Inequalities)	Relevant IF assistive-technology application is validated	Requires user studies, not just benchmarks

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16. How to Make Impact Stronger

In priority order:

#	Strategy	Why It Matters
1	**Demonstrate genuine performance/risk advantage**	A reliable result > a complicated architecture
2	**Release reproducible experiments**	Corruption generators, configs, eval scripts, model checkpoints
3	**Test generalisation**	Show gains persist across datasets, severities, unseen combinations
4	**Make decisions explainable**	Display failures, interventions, uncertainty — without claiming perfect causal explanation
5	**Validate one application**	Work with a real user group before claiming safety/accessibility benefits
6	**Create MS-Bench**	A reusable benchmark with {Missing, Noise, Corruption, Conflict, Manipulation, Compound} at controlled severities

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17. Comparison with Others & How We Stack Up

Direct Comparison Targets

vs. Method	What We Must Demonstrate	If Successful
Fixed Fusion	Better degradation resilience	Shows the value of adapting to input quality

DMS / Scalar Weighting	Better risk or robustness under compound failure	Supports the central policy hypothesis
PRIME	Better decisions where *different interventions* are needed	Establishes distinction from diagnosis + restoration
MCCF	Complementary value beyond missingness calibration	Shows action routing adds beyond calibrated uncertainty
Conflict-Focused Methods	Better handling of valid divergence vs. genuine failure	Supports semantic-diagnosis component
Selective Prediction Methods	Better risk-coverage trade-off	Supports deferral policy

The Expected Comparison Table (Illustrative)

```
Clean Noise Missing Conflict Compound Baseline 94 73 62 55 39 Adaptive baseline 95 82 76 70
58 MULTI-SHIELD 95 90 87 85 81
```

> **NOTE:** These are illustrative numbers, NOT expected results. The real result determines whether the hypothesis survives.

Three Possible Outcomes

Outcome	What It Means	Response
Best Case	Policy routing produces statistically significant improvements in selective risk + compound robustness, retains clean performance, generalises cross-dataset	**Strong research contribution.**
Mixed Result	Policy helps only for particular failure combinations or cost constraints	Narrow the contribution to those conditions; explain when routing is useful.
Negative Result	Strong scalar baseline performs equally or better	Simplify model; investigate *why*; reconsider research question. **This is NOT a failed project — an unsupported novelty claim is the greater risk.**

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18. Development Roadmap (What Needs to Be Done)

Phase-by-Phase Execution Plan

Detailed Phase Breakdown

Phase 0 — Literature Attack (Weeks 1–3)

> **DO NOT BUILD THE FINAL SYSTEM YET.**

Task	Deliverable
Read and dissect PRIME (2026) and DMS (2025)	Structured comparison matrix
Map 8 literature families (fusion, missing, noise, conflict, reliability, adaptive, selective, hallucination)	Literature family matrix
Determine what MULTI-SHIELD can contribute that prior art doesn't	Finalized research gap statement
Run Phase 0 Kill Check	Go/No-Go decision

Phase 1 — Data Pipeline (Weeks 3–5)

Task	Deliverable
Load CMU-MOSI, CMU-MOSEI, MELD	Dataset loaders
Align modalities (text, audio, visual)	Alignment pipeline
Validate data integrity (missing values, sync, label quality, duplicates, train/test leakage)	Data audit report
Create reproducible train/validation/test splits	Split configuration

Phase 2 — Baselines (Weeks 5–8)

Task	Deliverable
Implement Early Fusion	Baseline 1 results
Implement Late Fusion	Baseline 2 results
Implement Attention Fusion	Baseline 3 results
Implement strong recent adaptive method (DMS-style)	Baseline 4 results
Establish Macro-F1, accuracy, calibration, runtime on clean data	Baseline benchmark report

Phase 3 — Failure Generator (Weeks 7–10)

Task	Deliverable

Build modality missing generator	Missing modality pipeline
Build noise injection (Gaussian, progressive 10%→90%)	Noise pipeline
Build corruption injection (occlusion, blur, compression)	Corruption pipeline
Build conflict injection (swap modality labels)	Conflict pipeline
Build compound failure combinations (2-way, 3-way)	Compound pipeline
Create metadata records (sample ID, modality, degradation type, severity, ground truth)	Degradation metadata schema

Phase 4 — Proposed Mechanism (Weeks 10–16)

Task	Deliverable
Implement Reliability Engine (confidence, uncertainty, agreement, quality, failure indicators)	Evidence assessment module
Implement Failure-State Estimator (probabilistic classifier over failure taxonomy)	Diagnosis module
Implement Policy Router (maps failure state → action selection)	Policy router module
Implement individual actions (REWEIGHT, SUPPRESS, DEFER)	Action modules
Implement REPAIR (cross-modal reconstruction + reassessment) — Phase 2 only	Repair module (conditional)
Integration testing	End-to-end MULTI-SHIELD prototype

Phase 5 — Controlled Experiments (Months 4–6)

Full experimental suite:

Experiment	Purpose
E0 — Dataset integrity	Validate before any ML
E1 — Normal condition (clean data)	Ensure no clean-data sacrifice
E2 — Missing modality	Test each modality missing, pairs missing
E3 — Noise (10%→90% progressive)	Determine degradation curve
E4 — Cross-modal contradiction	Measure conflict detection + prediction + abstention

E5 — Semantic corruption	Valid input, wrong semantic content
E6 — Manipulated modality	Can system isolate suspicious evidence?
E7 — Compound failure (THE KILLER EXPERIMENT)	Image 30% corrupt + Audio missing + Text contradicts
E8 — Abstention	All modalities unreliable → system should defer
E9 — Severity analysis	At what level does system stop being trustworthy?
E10 — Ablation	Remove one component at a time → measure contribution
E11 — Cross-dataset generalisation	General principles vs. memorized corruption patterns

Phase 6 — Real-World Prototype (Months 6–8)

- Choose ONE practical domain (assistive tech, industrial monitoring, or smart city)
- Validate with naturally occurring degradation where possible
- Build interactive demo dashboard

Phase 7–9 — Papers, Final System, Demo (Months 7–12)

Phase	Deliverable
Paper 1	"Failure-State-Conditioned Decision Policies for Robust Multimodal Learning"
Paper 2 (conditional)	"Evaluating Multimodal Failure Diagnosis under Compound Degradation and Semantic Divergence"
Final Demo	Interactive dashboard with corruption sliders, real-time reliability visualization, baseline comparison

Team Division (4 People)

Person	Responsibilities
Person 1 — Research + Multimodal Modeling	Literature, baseline architectures, encoders, fusion
Person 2 — Reliability/Diagnosis	Corruption generator, reliability estimation, conflict detection, uncertainty
Person 3 — Evaluation	Benchmarks, experiments, ablations, statistical analysis, reproducibility
Person 4 — System + Visualization	API, dashboard, simulation, experiment controls, deployment

Four-Review College Survival Strategy

Review	Show	Don't Show
Review 1	Problem + literature + gap hypothesis	"We finished MULTI-SHIELD"
Review 2	Baseline + failure benchmark: "Existing methods degrade under controlled failures"	Final architecture claims
Review 3	MULTI-SHIELD prototype + preliminary results: "Our mechanism changes the degradation curve"	Unvalidated novelty claims
Review 4	Complete validation: benchmark + ablation + statistical analysis + real-world demo + limitations + papers	Everything omitted earlier

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19. Master Comparison Matrix: All Approaches

This is the comprehensive compare-and-contrast of every approach discussed:

Dimension 1: What Each Approach Diagnoses

Approach	Diagnoses Failure Type?	Diagnoses Severity?	Diagnoses Compound Failures?	Distinguishes Conflict from Corruption?
Fixed Fusion	■	■	■	■
Adaptive Weighting (DMS)	■ (uses confidence as proxy)	Partially (via weight magnitude)	■ (treats independently)	■
Missing-Modality Learning	Only missing	■	■	■
Uncertainty-Aware (AISTATS '25)	■ (uncertainty ≠ diagnosis)	■	■	Partially
PRIME (2026)	■ (prototype-based)	Partially	■ (focuses on intent recognition)	■
MMMC (ICML '25)	Only conflict	■	■	■
Sensor Expert Fusion (CVPR '25)	Only sensor dropout	■	■	■
MULTI-SHIELD	**■ (full taxonomy)**	**■ (progressive)**	**■ (core focus)**	**■ (research target)**

Dimension 2: What Actions Each Can Take

Approach	Reweight	Suppress	Repair	Defer/Abstain
Fixed Fusion	■	■	■	■
Adaptive Weighting (DMS)	■ (continuous)	Implicit (weight→0)	■	■
Missing-Modality Learning	■	Implicit	Sometimes (imputation)	■
Uncertainty-Aware	■ (uncertainty-based)	■	■	Partially
PRIME (2026)	■	■	■ (restoration)	■
MCCF (2026)	■	■	■	■ (conformal)
MULTI-SHIELD	■	■	■	■

Dimension 3: Evaluation Capabilities

Approach	Clean Performance	Robustness Curves	Calibration	Selective Risk	Failure Diagnosis Metrics	Compound Testing
Fixed Fusion	■	■	■	■	■	■
DMS	■	Partial	■	■	■	■
PRIME	■	■	■	■	Partial	■
MCCF	■	■	■	■	■	■
MULTI-SHIELD	■	■	■	■	■	■

Dimension 4: Core Philosophy Comparison

Approach	Core Question	Mechanism	Weakness
Fixed Fusion	"How to combine all modalities?"	Static architecture	Zero adaptability
Adaptive Weighting	"How much should each modality contribute?"	Continuous scalar weights	Doesn't know *why* to reweight

Missing-Modality	"What to do when input is absent?"	Imputation / dropout training	Only handles one failure type
Uncertainty-Aware	"How uncertain is the prediction?"	Evidential/Bayesian estimates	Uncertainty ≠ actionable diagnosis
Restoration (PRIME)	"Can we fix the damaged input?"	Prototype-conditioned reconstruction	Repair may introduce hallucinations
Conflict-Aware (MMMC)	"Do modalities agree?"	Conflict detection	Agreement ≠ reliability
MULTI-SHIELD	"What went wrong, and what should we do about it?"	Diagnosis → Policy → Action	Complexity; cascading diagnosis errors

Final Verdict: Strengths & Weaknesses Summary

Criterion	MULTI-SHIELD Strength	MULTI-SHIELD Weakness
Comprehensiveness	Only framework addressing full failure taxonomy + compound failures + selective prediction	Broad scope risks incomplete implementation
Actionability	Discrete intervention selection (Reweight/Suppress/Repair/Defer)	Policy router adds complexity and error propagation risk
Scientific Rigour	Pre-registered kill conditions, ablation plans, failure-boundary analysis	Unproven until experiments validate hypothesis
Practical Value	Explainable audit trail for every decision	Dashboard may consume development time
Research Novelty	Failure-state-conditioned policy under compound degradation	PRIME (2026) is close; must demonstrate clear delta
Generalisability	Domain-agnostic design	Synthetic corruption may not transfer to real degradation

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IMPORTANT:

The Next Move

Do not start coding.

> The next move is the **Phase 0 Literature-Gap Attack**: 1. Build a structured matrix of the closest recent methods across: missing modality, noisy modality, conflict, uncertainty, adaptive fusion, restoration, abstention, compound degradation 2. Determine exactly what MULTI-SHIELD can contribute that those systems do not already provide 3. Only after that: freeze the architecture, research questions, and final title

> *The dashboard, fancy architecture diagram, and name are secondary. The research loop is the actual project.*